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Will SpaceX Falcon 9 Stick the Landing?

A predictive-analytics capstone forecasting first-stage booster recovery from public launch data

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Data Science Capstone Project

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GitHub: https://github.com/ejnaik/Data-Science-Specialization/tree/main/Applied_DataScience_Capstone

Outline

1. Executive Summary
2. Introduction
3. Methodology
4. Results
5. Conclusion
6. Appendix

From Raw Launch Data to a Landing Predictor

Summary of Methodologies

- Data collection — SpaceX REST API + Wikipedia web scraping
- Data wrangling — cleaning, imputation, outcome labelling
- EDA with data visualization (Matplotlib/Seaborn)
- EDA with SQL (IBM Db2)
- Interactive visual analytics — Folium map + Plotly Dash
- Predictive analysis (classification) — 4 tuned models

Summary of All Results

90
Falcon 9 launches analysed (2010–2020)

3
Launch sites profiled
CCAFS · KSC · VAFB

87.8%
Best full-dataset classification accuracy

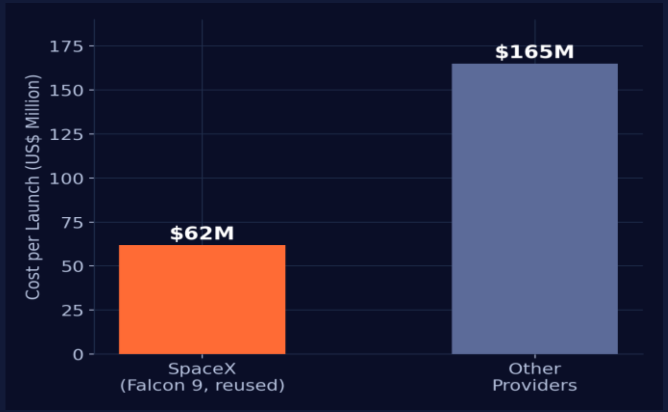
- Success rate climbed from 0% (2013) to ~90% (2019–20)
- KSC LC-39A leads all pads with a 76.9% success ratio
- Tuned SVM gives the strongest overall fit (87.8% accuracy)
- A reusable first stage is the core driver of SpaceX's ~2.7x cost advantage over competitors

INTRODUCTION

Why Predict a Falcon 9 Landing?

Project Background and Context

SpaceX advertises Falcon 9 launches at US$62M — other providers charge upward of US$165M. Much of that saving comes from reusing the rocket's first stage. If we can predict whether the first stage will land, we can estimate the true cost of a launch, and give a competing bidder or investor a data-driven view of SpaceX's economics.



Questions to Be Answered

- How do payload mass, launch site, flight count & orbit affect landing success?
- Does the landing success rate increase over the years?
- What is the best algorithm for binary classification in this case?

SECTION 1

Methodology

How we collected, cleaned, explored, and modelled the data

METHODOLOGY

A Six-Stage Analytics Pipeline

1

Data Collection

SpaceX API + Wikipedia scraping

2

Data Wrangling

Cleaning, imputation, labelling

3

EDA & Viz

Matplotlib / Seaborn charts

4

EDA with SQL

IBM Db2 queries

5

Visual Analytics

Folium map + Plotly Dash

6

Predictive Modelling

4 tuned classifiers

Data Collection

Describe How Data Were Collected

Two complementary data-collection methods were combined to obtain complete Falcon 9 launch records: requests to the SpaceX REST API for structured mission telemetry, and web scraping of the Falcon 9 launch-history table on Wikipedia for cross-referenced historical fields. Both sources were reconciled into a single, analysis-ready DataFrame.

SpaceX REST API

FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, Serial, Latitude/Longitude

Wikipedia Web Scraping

Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

Data Collection – SpaceX API

1. Request rocket-launch data from SpaceX API
2. Decode .json() response → pd.json_normalize()
3. Request booster/payload/launch-pad details via helper functions
4. Assemble fields into a dictionary
5. Build a DataFrame from the dictionary
6. Filter to Falcon 9 only
7. Impute missing PayloadMass (.mean())
8. Export cleaned table to CSV

Data Collection – Web Scraping

1. Request Falcon 9 launch-history page from Wikipedia
2. Parse HTML response into a BeautifulSoup object
3. Extract column names from the HTML table header
4. Parse each table row into a structured record
5. Assemble records into a dictionary
6. Build a DataFrame from the dictionary
7. Export the scraped table to CSV

Cleaning the Data & Defining the Target Label

From Landing Outcome to Binary Class

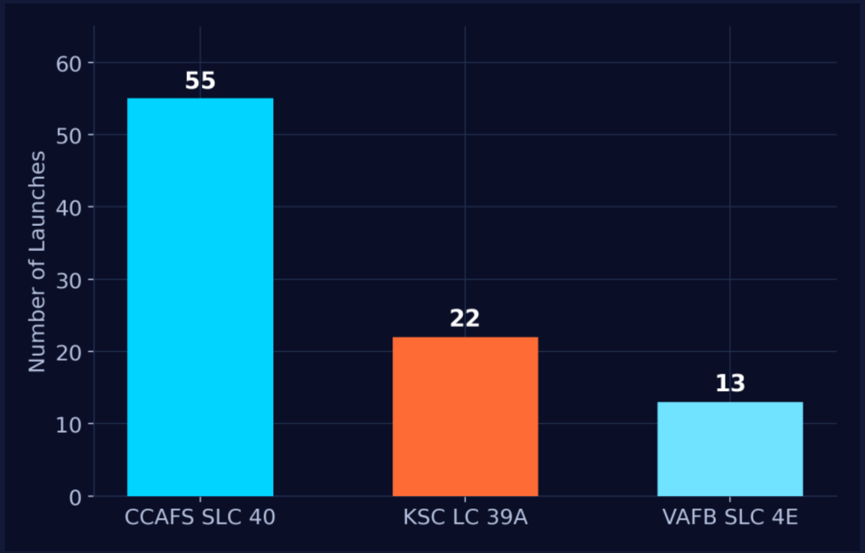
Raw outcomes describe where and how a booster landed — e.g. True Ocean / False Ocean, True RTLS / False RTLS, True ASDS / False ASDS. "True" outcomes are a controlled, successful landing to a target zone; "False" outcomes are a failed attempt.

Class = 1 → booster successfully landed

Class = 0 → landing attempt was unsuccessful

- Calculate launches per site and per orbit
- Tabulate mission-outcome counts per orbit type
- Derive the landing_outcome label from Outcome
- One-Hot Encode categorical features for modelling

Falcon 9 Launches Analysed



90 Falcon 9 missions (2010–2020) across 3 pads: CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.

EDA with Data Visualization

Charts Plotted & Why

Flight Number vs. Payload Mass · Flight Number vs. Launch Site · Payload Mass vs. Launch Site · Orbit Type vs. Success Rate · Flight Number vs. Orbit Type · Payload Mass vs. Orbit Type · Success Rate Yearly Trend

Scatter Plots

Show the relationship between two continuous variables — if a relationship exists, it could be used as a model feature.

Bar Charts

Compare a measured value (success rate) across discrete categories such as launch site or orbit type.

Line Chart

Show the trend in landing-success rate over time (a time series), revealing the program's learning curve.

EDA with SQL

Ten Analytical Queries Against SPACEXDATASET (IBM Db2)

1. Unique launch site names
2. Records where site begins with 'CCA'
3. Total payload mass by customer (NASA CRS)
4. Average payload mass by booster version
5. Date of first successful ground-pad landing
6. Boosters with drone-ship success in a payload band
7. Total successful vs. failed mission outcomes
8. Booster versions carrying maximum payload
9. Failed drone-ship landings in 2015
10. Ranked landing-outcome counts within a date range

Build an Interactive Map with Folium

Map Objects Used

Markers

A Circle, Popup label, and Text label for NASA Johnson Space Center and every launch site, positioned by latitude/longitude.

MarkerCluster

Colour-coded markers (green = success, red = failure) at each site to reveal relative site success rates.

PolyLines

Distance measurements from a representative site (KSC LC-39A) to nearby railway, highway, coastline and city.

Build a Dashboard with Plotly Dash

Plots & Interactions Added

Site Dropdown

Filter every chart to All Sites or one specific launch pad.

Success Pie Chart

Total successes per site, or Success vs. Fail split for one selected site.

Payload Slider

Drag to bound the payload-mass range (kg) shown across all charts.

Payload Scatter

Payload mass vs. landing success, coloured by booster version category.

Predictive Analysis (Classification)

1. Extract Class column as NumPy array (target Y)
2. Standardize features with StandardScaler
3. 80/20 train-test split (train_test_split)
4. GridSearchCV (cv = 10) over 4 algorithms
5. Score each tuned model with .score()
6. Inspect confusion matrices
7. **Compare Jaccard & F1 to pick a winner**

Four Algorithms, Grid-Searched

Logistic Regression

tuned on: C, penalty, solver

Support Vector Machine

tuned on: C, gamma, kernel

Decision Tree

tuned on: criterion, max_depth, splitter

K-Nearest Neighbours

tuned on: n_neighbors, p, algorithm

RESULTS

What the Data Told Us

EDA visuals · SQL insights · interactive map & dashboard · predictive scores

SECTION 2

EDA Results

Visualization findings and SQL query results

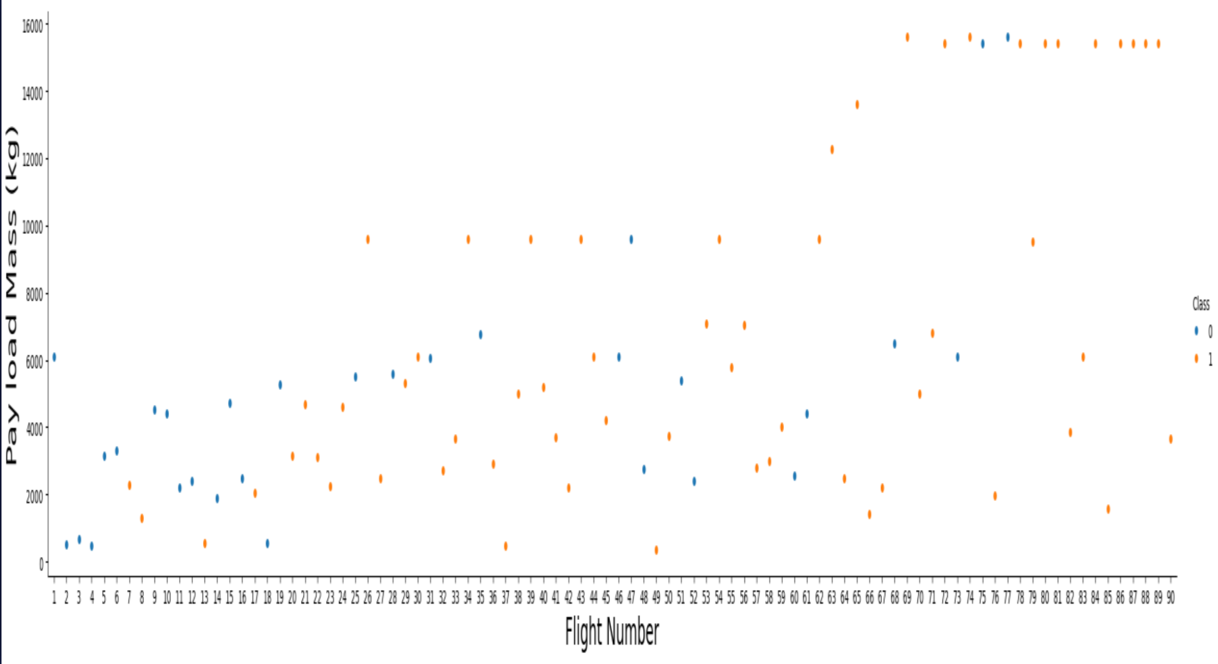
Flight Number vs. Launch Site

What We Plotted

A scatter plot of Flight Number against Launch Site, coloured by landing outcome (Class).

Explanation

- The earliest flights all failed while the latest flights all succeeded.
- CCAFS SLC-40 hosts roughly half of all launches.
- VAFB SLC-4E and KSC LC-39A show higher success rates.
- Each new launch trends toward a higher rate of success.



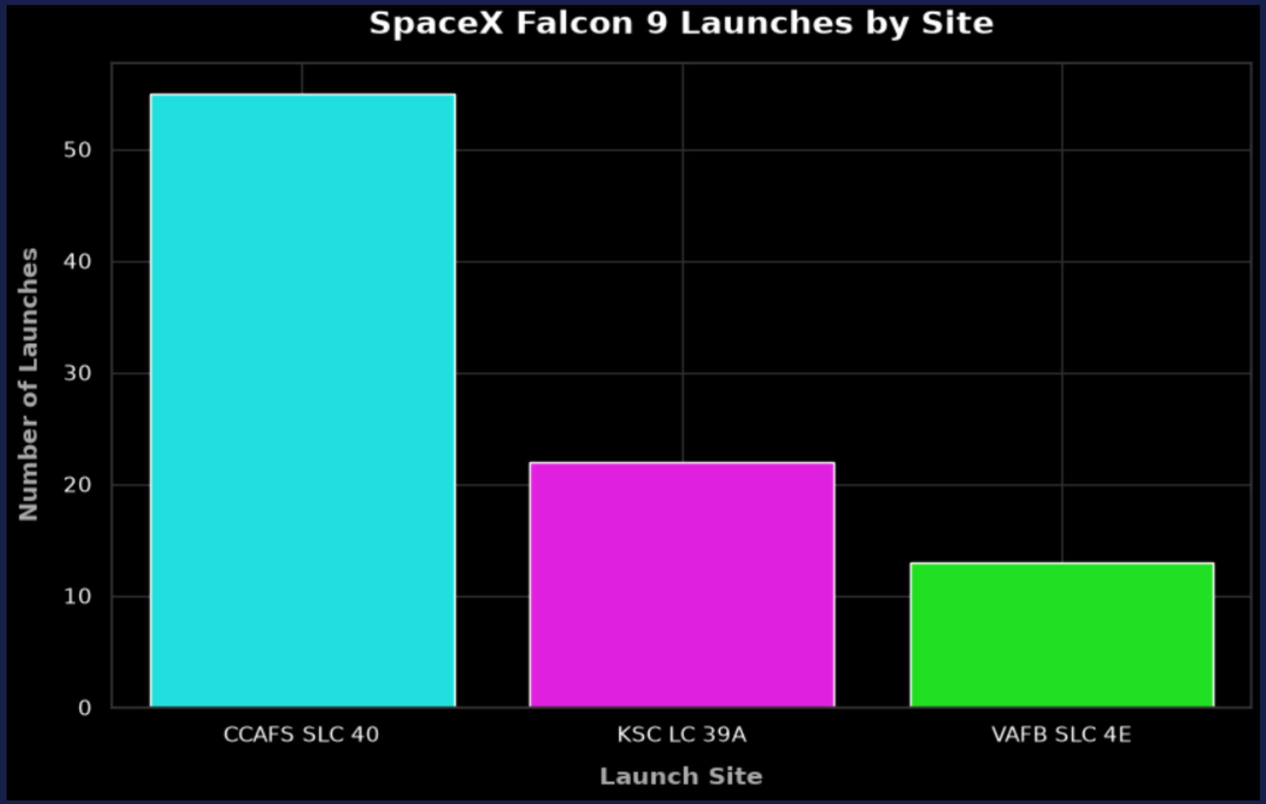
Payload vs. Launch Site

What We Plotted

A scatter plot of Payload Mass against Launch Site, coloured by landing outcome (Class).

Explanation

- For every launch site, higher payload mass associates with a higher success rate.
- Most launches with payload mass over 7,000 kg were successful.
- KSC LC-39A shows a 100% success rate even for payload under 5,500 kg.



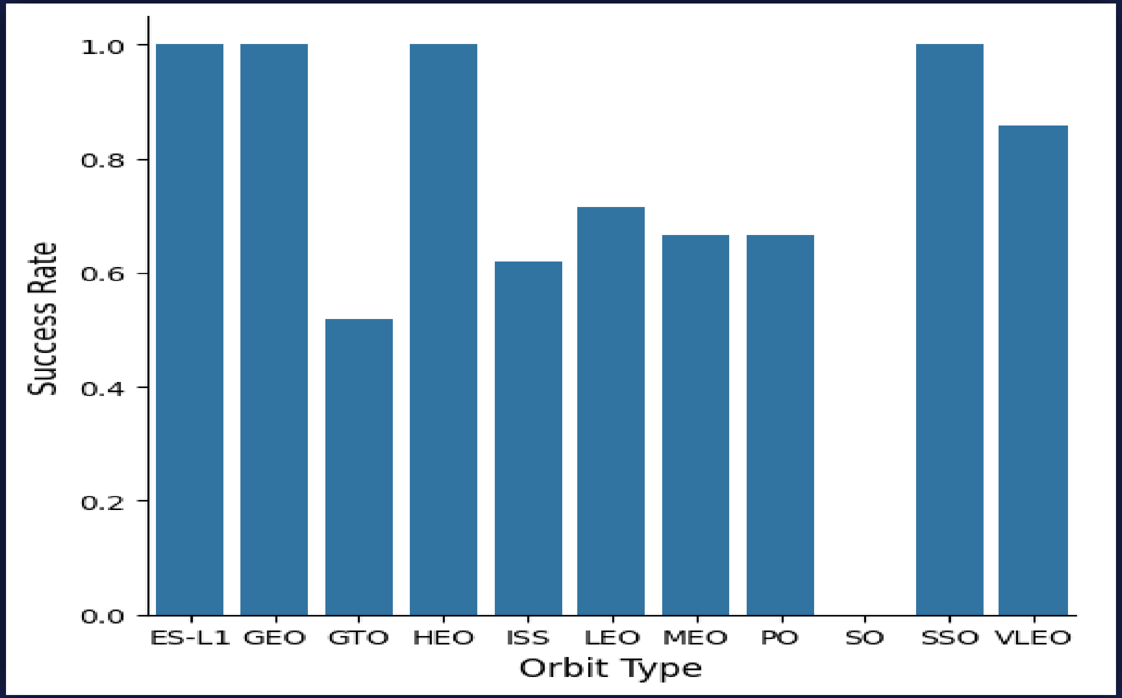
Success Rate vs. Orbit Type

What We Plotted

A bar chart of the success rate for each orbit type present in the dataset.

Explanation

- ES-L1, GEO, HEO & SSO: 100% success rate.
- SO: 0% success rate (single mission).
- GTO, ISS, LEO, MEO & PO: 50–85% success rate.



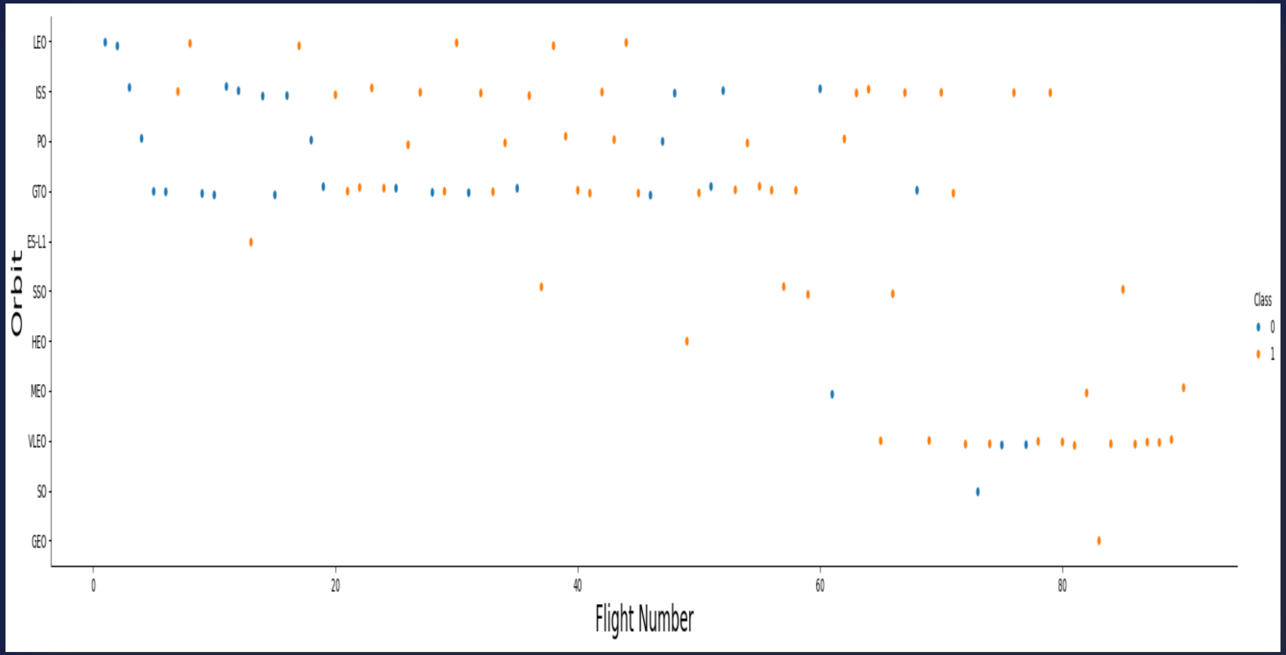
Flight Number vs. Orbit Type

What We Plotted

A scatter plot of Flight Number against Orbit Type, coloured by landing outcome (Class).

Explanation

- In the LEO orbit, success appears related to the number of flights.
- No clear relationship is visible between flight number and success in the GTO orbit.



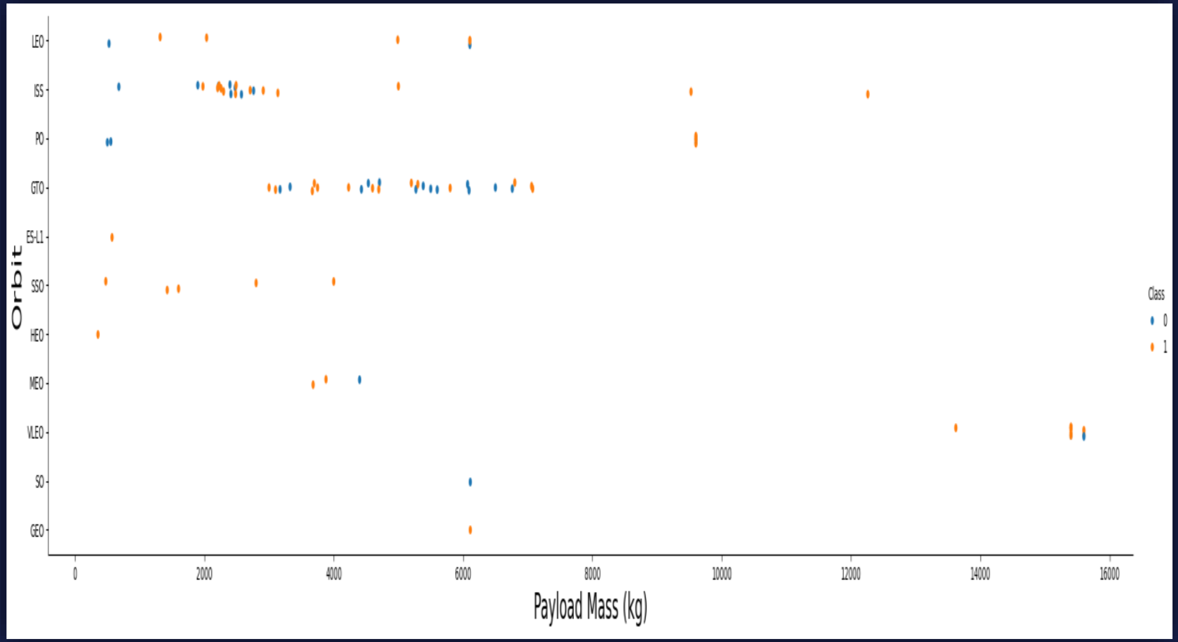
Payload vs. Orbit Type

What We Plotted

A scatter plot of Payload Mass against Orbit Type, coloured by landing outcome (Class).

Explanation

- Heavy payloads have a negative influence on GTO orbit outcomes.
- Heavy payloads have a positive influence on Polar LEO (ISS) orbit outcomes.



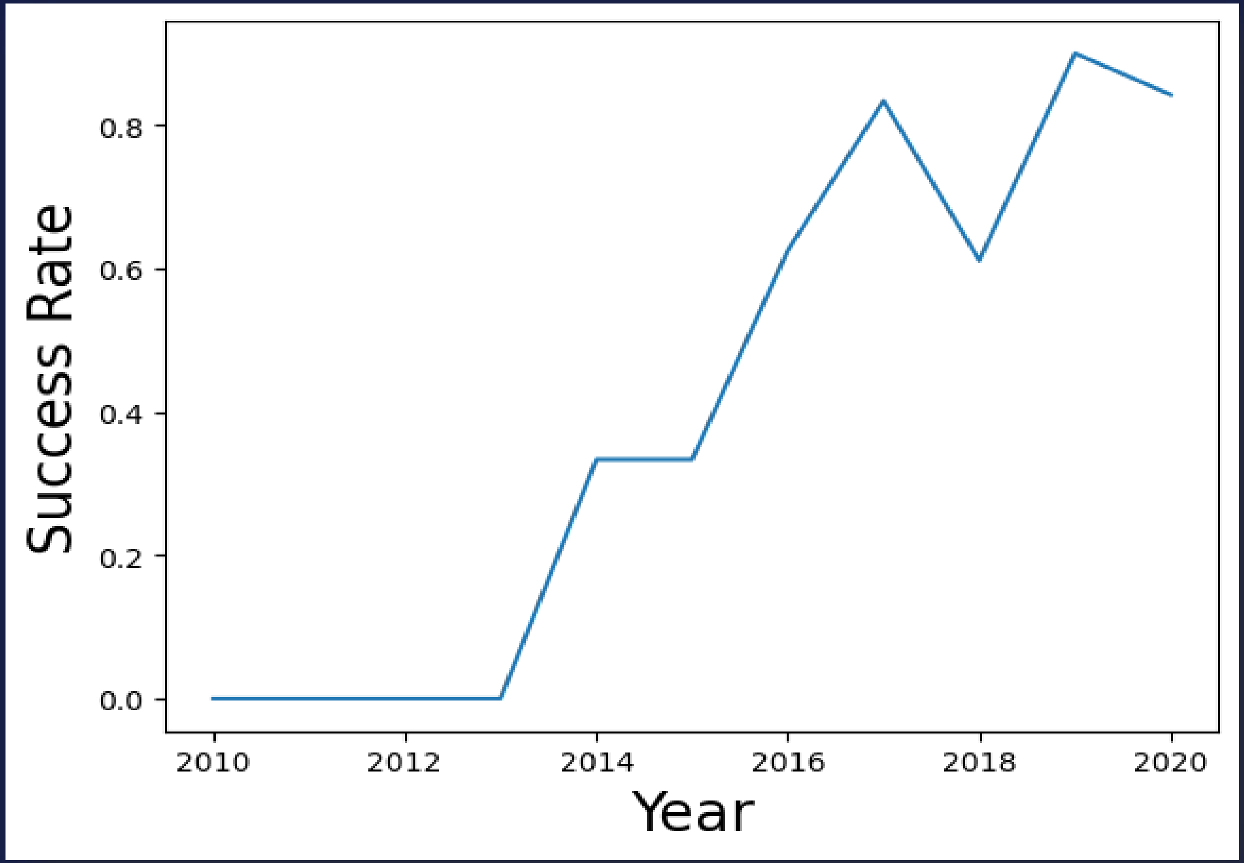
Launch Success Yearly Trend

What We Plotted

A line chart of the average landing-success rate for each year in the dataset.

Explanation

- The success rate since 2013 kept increasing through 2020.
- This reflects the rapid maturation of the reusable-booster program.



All Launch Site Names

```sql
SELECT DISTINCT launch_site FROM SPACEXDATASET;
```

QUERY GOAL

Display the names of the unique launch sites used in the SpaceX mission dataset.

RESULT

- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

Launch Site Names Begin with 'CCA'

```
SELECT * FROM SPACEXDATASET
WHERE launch_site LIKE 'CCA%' LIMIT 5;
```

QUERY GOAL

Display 5 records where the launch site begins with the string 'CCA'.

RESULT

5 records

Earliest CCAFS LC-40 missions (2010–2013), including the Dragon Spacecraft Qualification Unit and the first two Dragon/CRS demo flights.

Total Payload Mass

```
SELECT SUM(payload_mass__kg_) AS total_payload_mass
FROM SPACEXDATASET WHERE customer = 'NASA (CRS)';
```

QUERY GOAL

Calculate the total payload mass carried by boosters launched for NASA (CRS).

RESULT

45,596 kg

Average Payload Mass by F9 v1.1

```
SELECT AVG(payload_mass__kg_) AS average_payload_mass
FROM SPACEXDATASET WHERE booster_version LIKE '%F9 v1.1%';
```

QUERY GOAL

Calculate the average payload mass carried by booster version F9 v1.1.

RESULT

2,534 kg

First Successful Ground Landing Date

```sql
SELECT MIN(date) AS first_successful_landing
FROM SPACEXDATASET WHERE landing__outcome = 'Success (ground pad)';
```

QUERY GOAL

Find the date of the first successful landing outcome on a ground pad.

RESULT

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT booster_version FROM SPACEXDATASET
WHERE landing__outcome = 'Success (drone ship)'
  AND payload_mass__kg_ BETWEEN 4000 AND 6000;
```

QUERY GOAL

List the booster names that successfully landed on a drone ship with a payload mass between 4,000 kg and 6,000 kg.

RESULT

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
SELECT mission_outcome, COUNT(*) AS total_number
FROM SPACEXDATASET GROUP BY mission_outcome;
```

QUERY GOAL

Calculate the total number of successful and failed mission outcomes.

RESULT

- Success — 99
- Failure (in flight) — 1
- Success (payload status unclear) — 1

Boosters Carried Maximum Payload

```
SELECT booster_version FROM SPACEXDATASET
WHERE payload_mass__kg_ = (SELECT MAX(payload_mass__kg_) FROM SPACEXDATASET);
```

QUERY GOAL

List the names of the booster versions that carried the maximum payload mass.

RESULT

11 boosters

All F9 B5 boosters tied at the maximum recorded payload mass (e.g. B1048, B1049, B1051, B1056, B1058, B1060 across multiple flights).

2015 Launch Records

```
SELECT MONTHNAME(date) AS month, date, booster_version,
  launch_site, landing__outcome FROM SPACEXDATASET
WHERE landing__outcome = 'Failure (drone ship)' AND YEAR(date) = 2015;
```

QUERY GOAL

List the failed landing outcomes on a drone ship, their booster versions, and launch site names for the months of the year 2015.

RESULT

- January 2015 — F9 v1.1 B1012 — CCAFS LC-40
- April 2015 — F9 v1.1 B1015 — CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT landing__outcome, COUNT(*) AS count_outcomes
FROM SPACEXDATASET WHERE date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY landing__outcome ORDER BY count_outcomes DESC;
```

QUERY GOAL

Rank the count of landing outcomes between the given dates in descending order.

RESULT

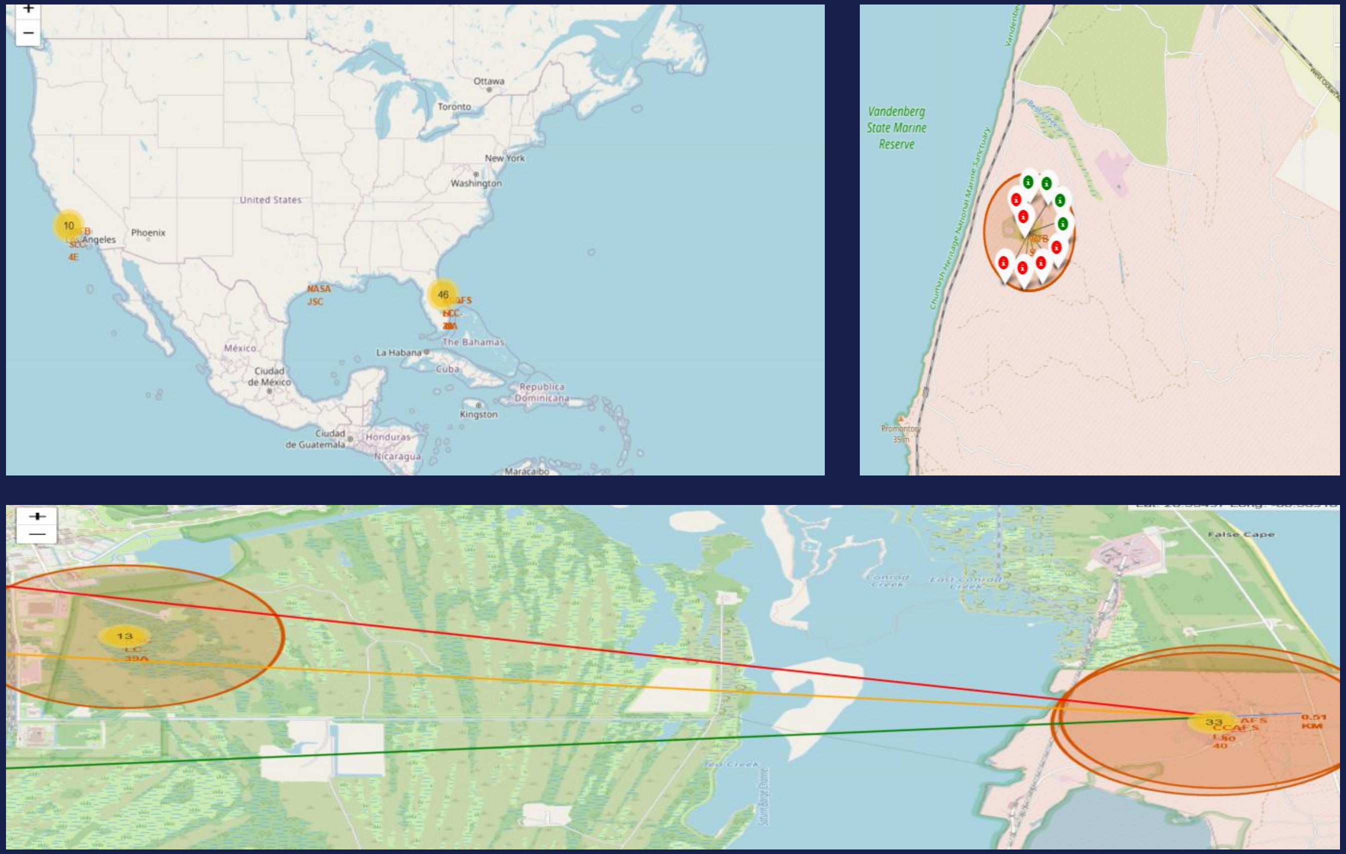
- No attempt — 10
- Failure (drone ship) — 5
- Success (drone ship) — 5
- Controlled (ocean) — 3
- Success (ground pad) — 3

SECTION 3

Interactive Map (Folium)

Geospatial patterns behind every launch site

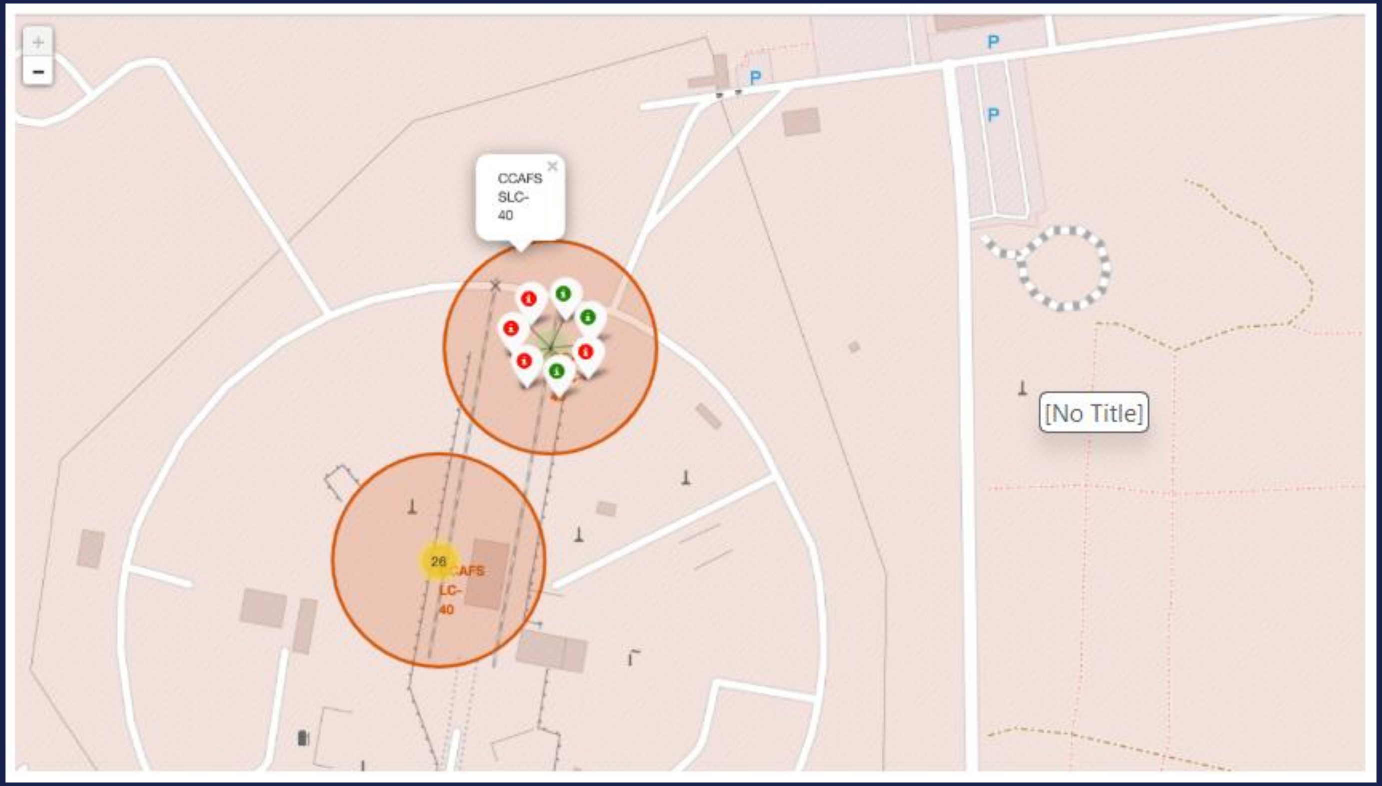
All Launch Sites' Location Markers on a Global Map



Findings

- Most launch sites are close to the Equator — Earth's surface there already moves at ~1,670 km/h, helping a rocket reach orbital velocity more efficiently.
- All launch sites are close to the coast, minimising the risk of debris from a failed launch falling near populated areas.

Colour-Labeled Launch Records on the Map



Findings

- Colour-coded markers make it easy to identify which launch sites have relatively high success rates.
- Green marker = successful launch.
- Red marker = failed launch.
- Launch site KSC LC-39A shows a very high success rate.

Distance from Launch Site KSC LC-39A to Its Proximities

Findings

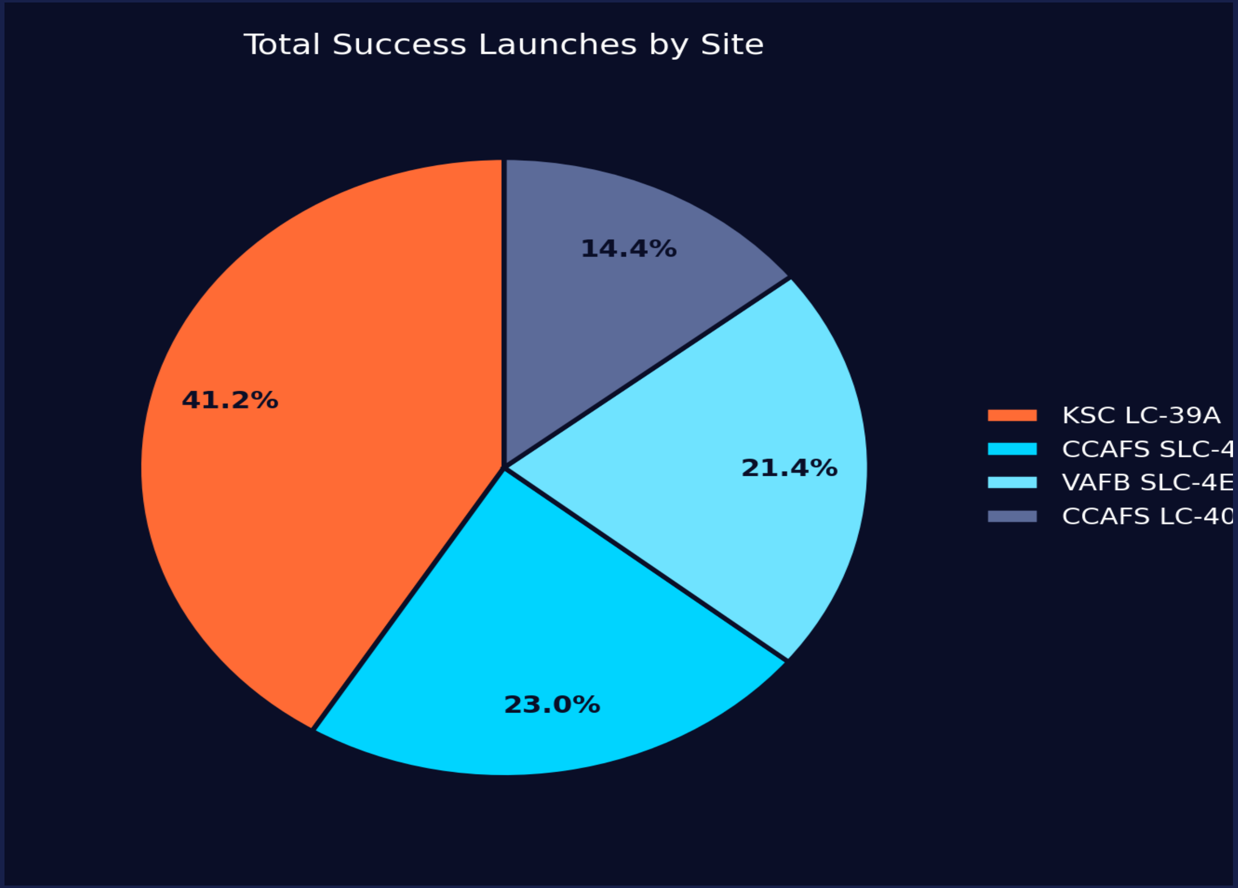
- Relatively close to railway (15.23 km).
- Relatively close to highway (20.28 km).
- Relatively close to coastline (14.99 km).
- Relatively close to its closest city, Titusville (16.32 km).
- A failed rocket at high speed can cover 15–20 km within seconds — a real safety consideration for populated areas.

SECTION 4

Dashboard (Plotly Dash)

Interactive exploration of success by site, payload, and booster

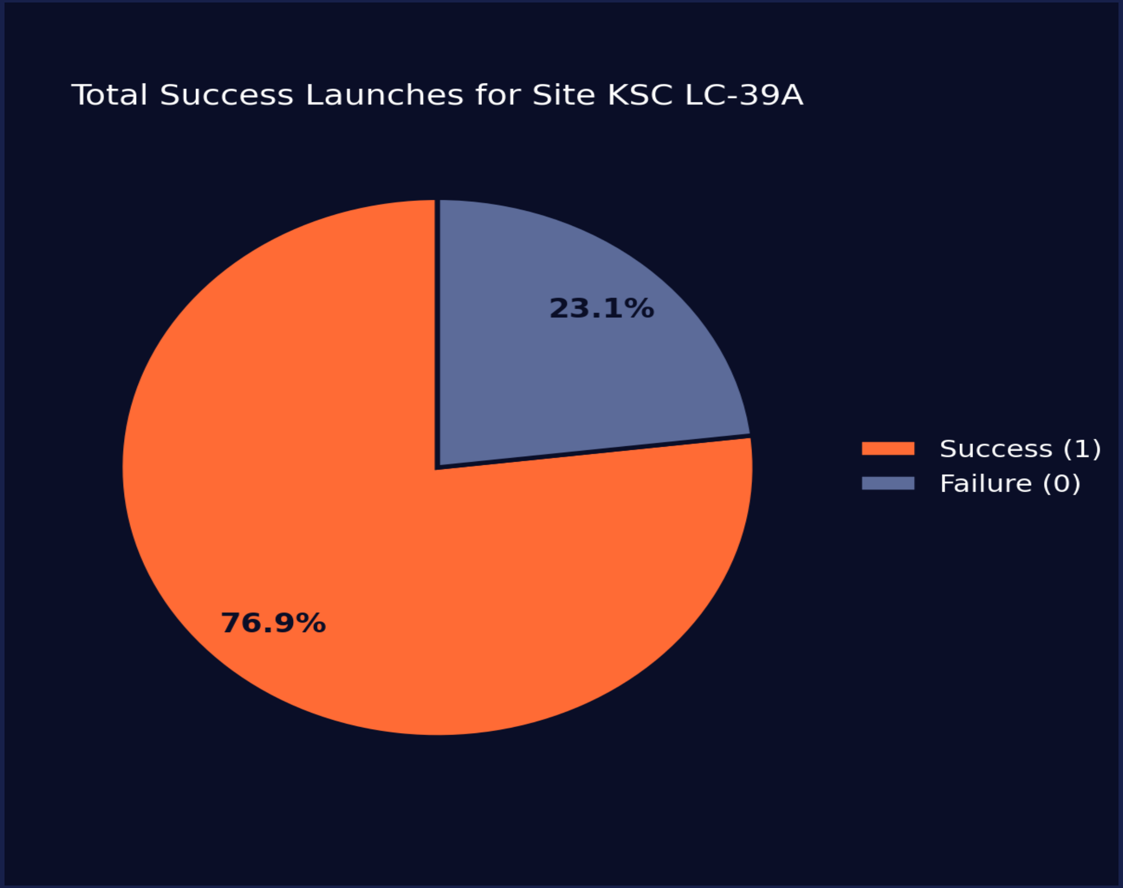
Launch Success Count for All Sites



Findings

- The pie chart shows that of all sites, KSC LC-39A has the most successful launches (41.2%).
- CCAFS SLC-40 follows at 23.0%, then VAFB SLC-4E at 21.4% and CCAFS LC-40 at 14.4%.

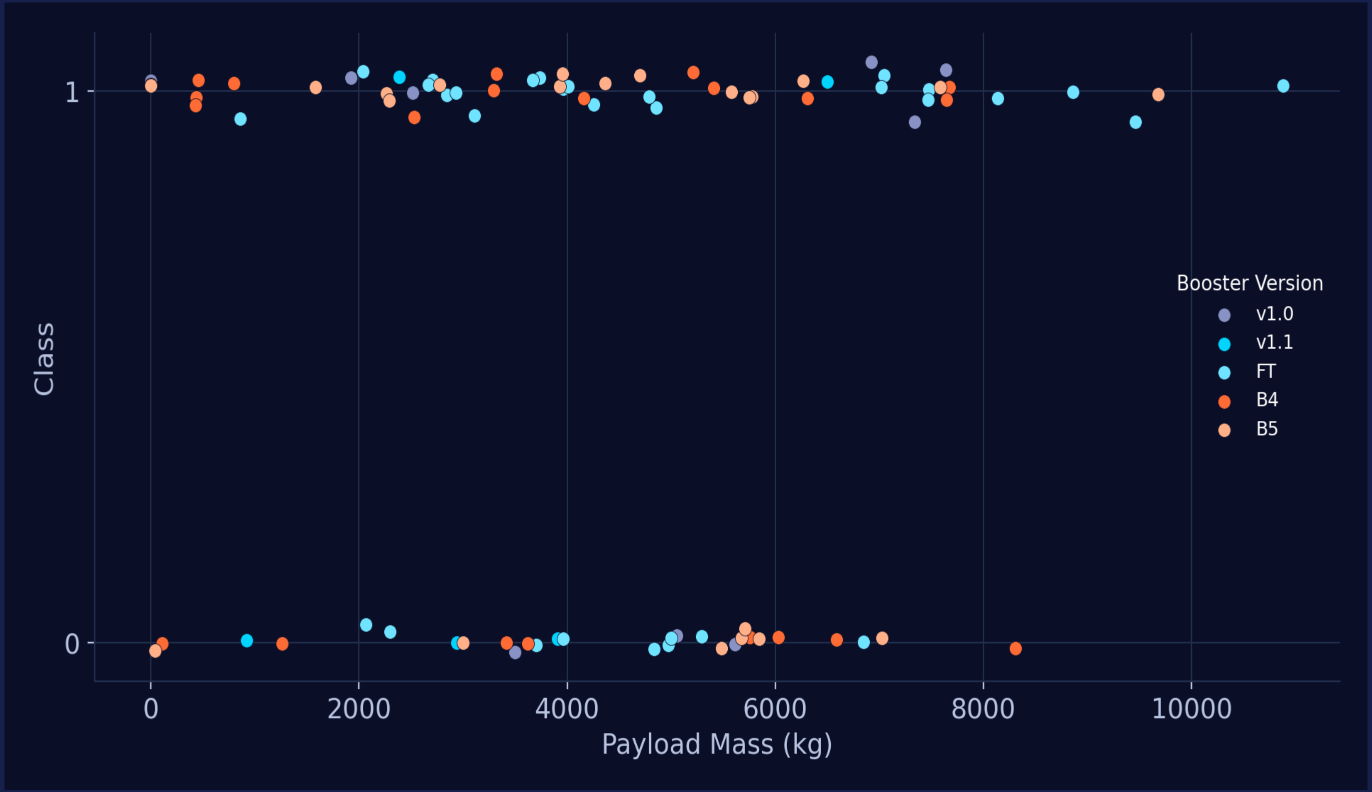
Launch Site with Highest Launch Success Ratio



Findings

- KSC LC-39A has the highest launch success rate of any pad: 76.9%.
- 10 successful landings vs. only 3 failed landings at this site.

Payload Mass vs. Launch Outcome for All Sites



Findings

- Filtering the payload-range slider shows payloads between roughly 2,000 and 5,500 kg have the highest concentration of successful landings.
- Newer booster versions (FT, B4, B5) dominate the higher-payload, higher-success region of the chart.

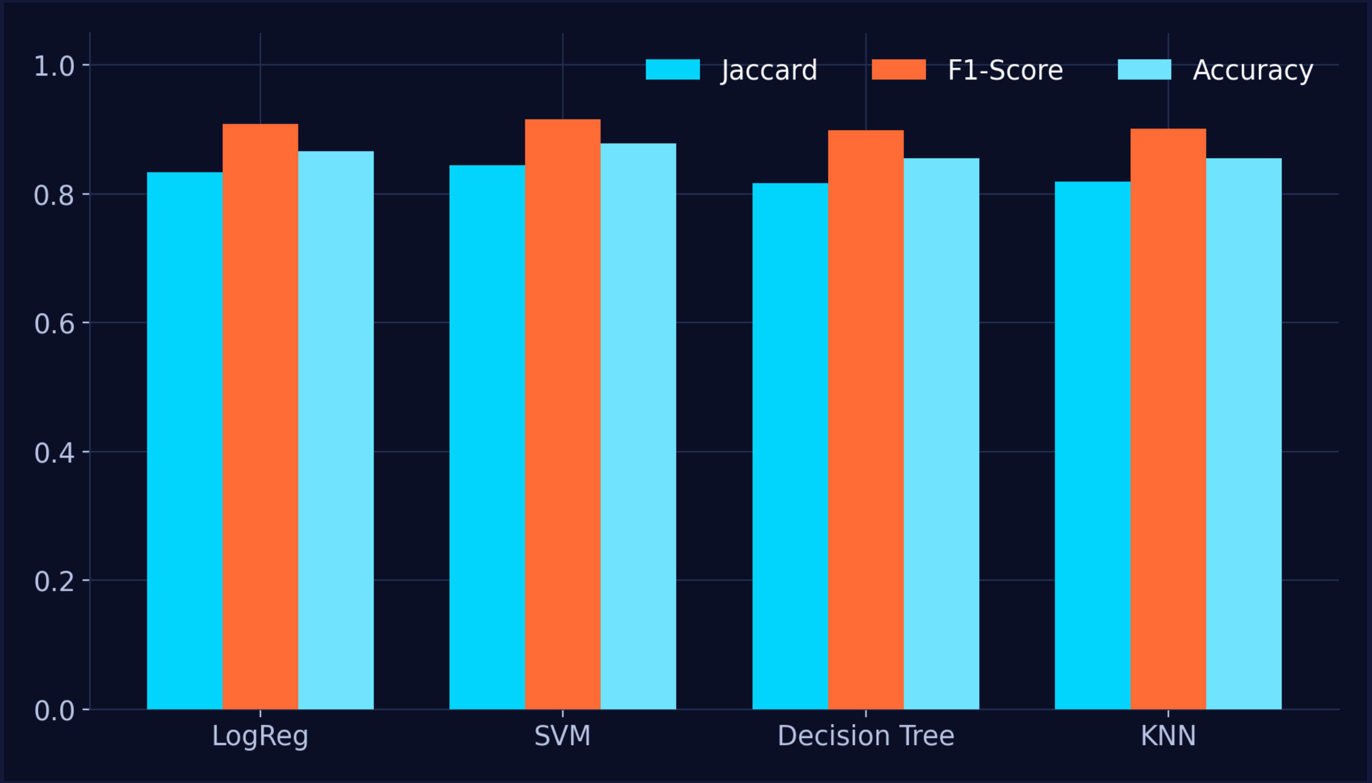
SECTION 5

Predictive Analysis

Which classifier best predicts a successful landing?

Classification Accuracy

Scores on the Full Dataset



Test-Set Snapshot

18 held-out launches (20% split)

LogReg

83.3% accuracy

SVM

83.3% accuracy

Decision Tree

83.3% accuracy

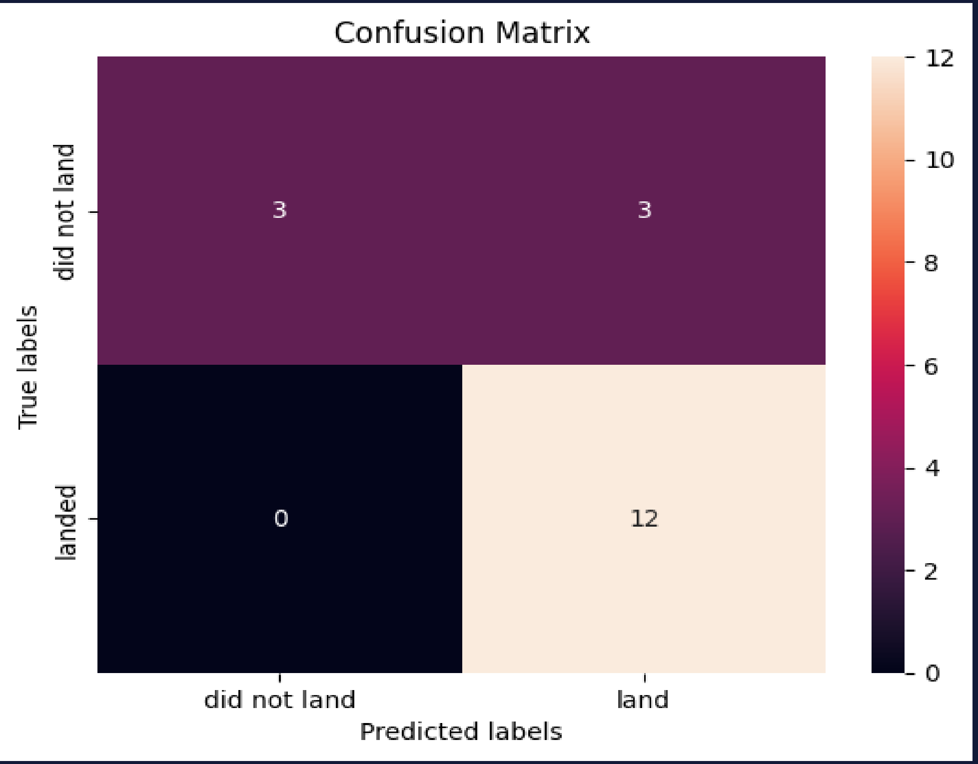
KNN

83.3% accuracy

All four models tie on this small 18-sample test split — differences only emerge when scored against the full 90-launch dataset.

Confusion Matrix

Logistic Regression — Test Set



What It Tells Us

- 12 landings correctly predicted, 3 non-landings correctly predicted
- 0 missed landings (no false negatives) — the model never fails to flag a successful landing
- 3 false positives — the main error mode is over-predicting success
- Best CV training accuracy: Decision Tree at 90.2% (max_depth = 8, gini criterion)
- Best full-dataset accuracy: tuned SVM at 87.8% (C = 1.0, RBF kernel)

Conclusions

- Falcon 9 first-stage landings are learnable: a tuned SVM classifies outcomes with 87.8% accuracy on the full dataset
- Low-to-moderate payload mass (2,000–5,500 kg) is associated with the strongest landing outcomes
- Nearly all launch sites are close to the Equator and to the coast — both a physics and a safety advantage
- Landing success has climbed steadily since 2013, reaching ~90% by 2019–2020
- KSC LC-39A is SpaceX's top-performing pad, both in volume and in success ratio (76.9%)
- Orbits ES-L1, GEO, HEO and SSO show a perfect 100% landing-success record

Beyond the Checklist: What Surprised Us

CV-winner ≠ dataset-winner

Decision Tree tops 10-fold cross-validation (90.2%), but SVM generalises best across the full dataset (87.8%) — CV rank and holdout rank can diverge on small (n = 90) samples, so we validate on multiple views before picking a production model.

Reusability is the real cost lever

At $62M vs. $165M+ per launch, a successful first-stage landing is worth roughly $100M in avoided hardware cost — turning a landing-outcome classifier into a lightweight pricing signal for a launch bid.

Geography is a hidden feature

Every pad sits near the Equator and the coast — an engineering constraint, not a modelling artefact — showing how domain context should guide feature selection, not just raw correlation.

Zero false negatives matters most

The tuned model never misses an actual landing (0 FN) — for a cost-estimation use case, that asymmetry is more valuable than raw accuracy, since underestimating cost is the costlier business mistake.

DELIVERABLES

GitHub Repository & Deliverables

GitHub Link: https://github.com/ejnaik/Data-Science-Specialization/tree/main/Applied_DataScience_Capstone

GitHub Repository

- 01 · Data Collection — SpaceX API notebook
- 02 · Data Collection — Web Scraping notebook
- 02b · Data Wrangling notebook
- 03 · EDA with SQL notebook
- 03b · EDA with Data Visualization notebook
- 04 · Interactive Map (Folium) notebook
- 05 · Plotly Dash application (spacex_dash_app.py)
- 06 · Machine-Learning Prediction notebook

Report & Presentation

- This slide deck, submitted as a PDF
- Full written report — A4 PDF, LaTeX source
- All chart & map assets regenerated from source data
- Reproducible: rerun each notebook end-to-end from raw data to results

APPENDIX

Hyperparameter Grids Used in GridSearchCV

Logistic Regression

C: [0.01, 0.1, 1] · penalty: [l2] · solver: [lbfgs]

SVM

C: log-spaced grid · gamma: log-spaced grid · kernel: rbf/linear/poly/sigmoid

Decision Tree

criterion: [gini, entropy] · max_depth: [2..18] · splitter, min_samples options

K-Nearest Neighbours

n_neighbors: [1..10] · algorithm: [auto, ball_tree, kd_tree, brute] · p: [1, 2]

GitHub https://github.com/ejnaik/Data-Science-Specialization/tree/main/Applied_DataScience_Capstone